Mahmoud Alyosify

NLP Engineer | Language Model Fine-Tuning, Text Understanding & Evaluation
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PROFESSIONAL SUMMARY

NLP Engineer working across the full language stack — **tokenisation and decoding, domain fine-tuning, information extraction, semantic search and evaluation**. Currently an **AI Engineer Intern at Injaz Tech**, building retrieval-augmented question answering with hybrid search, cross-encoder re-ranking and query rewriting. Master's thesis at **Queen's University, Canada** (expected **Nov 2026**, **Digilians Presidential Scholarship**) develops a training-free decoding method that makes LLMs generate **47.6% fewer tokens** with no answer cut off, evaluated on **MT-Bench, AlpacaEval and Arena-Hard-Auto** in English and six other languages with an **LLM judge** and **NLI-based content coverage**. Fine-tuned **Meta-Llama-3-8B** on a **159,826-record** domain corpus, and built an **entity-and-relation extraction pipeline** that converts unstructured open-source text into a queryable knowledge graph. Grounded in the part that decides whether NLP ships: evaluation that does not flatter the model.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Thesis (CISC 898):** *MinimalLM: Verbosity-Aware Decoding for Token-Efficient Language Model Inference* — Supervisor: Dr. Rahatara Ferdousi.
- Coursework: **Large Language Models, Generative AI, Deep Learning, Cloud Computing, End-to-End ML Pipelines**.

B.Sc. Computer & Information Science (Bioinformatics) — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

NLP Core	Transformers, Tokenisation, Text Summarisation, Extractive Question Answering, Information & Relation Extraction, Knowledge-Graph Construction, Text Classification, Semantic Similarity & Redundancy Detection, Sentence Embeddings, Semantic Search, RAG, Hybrid Search (BM25 + Dense, RRF), Cross-Encoder Re-Ranking, Query Rewriting
Language Models	Instruction Fine-Tuning, LoRA, QLoRA, PEFT, Custom Decoding Strategies, Prompt Engineering, Structured Outputs, Self-Hosted Inference · *Mistral-7B, Llama-3.1-8B, Meta-Llama-3-8B, Qwen2.5-3B*
Frameworks	PyTorch, Hugging Face Transformers / Datasets / PEFT, TensorFlow, Scikit-learn, NLTK, LangChain, LangGraph, sentence-transformers, FAISS, ChromaDB, Pandas, NumPy
Evaluation	BERTScore, ROUGE, LLM-as-Judge, NLI-Based Content Coverage, MT-Bench, AlpacaEval, Arena-Hard-Auto, Multilingual Evaluation, Non-Inferiority Testing, Ablation Studies, Reproducible Benchmarking
Deployment	AWS, Docker, FastAPI, Streamlit, ONNX, CI/CD, Git/GitHub, Linux, SQL
Programming	Python, SQL, C++, C#, JavaScript, R, Bash

PROFESSIONAL EXPERIENCE

AI Engineer Intern — *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) — Maadi, Cairo, Egypt

- Built RAG question answering: header-aware chunking, **FAISS**, BM25 + dense hybrid search, **cross-encoder re-ranking**, query rewriting and multi-turn follow-ups.
- Raised top-1 retrieval accuracy from **75% to 94%**; built an LLM-judge and key-fact evaluation suite run as a CI gate.

Part-Time Instructor — Machine Learning & AI for Business — *Jul 2025 – Present*
National Telecommunication Institute (NTI) — Egypt

- Teach **Machine Learning** (90-hour bootcamps) and **AI for Business** modules in NTI's Summer Internships Program, on separate engagements.

AI in Cybersecurity Intern — *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) — Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) — *Feb 2024 – Mar 2025*
National Company for Roads Building & Development — Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles**; co-developed an internal data-management application in **C#, .NET, SQL Server**.

Online Instructor — AI, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

SELECTED NLP PROJECTS

MinimalLM: Verbosity-Aware Decoding for Token-Efficient Language Model Inference *2026*
MSc Thesis (CISC 898), Queen's University · Supervisor: Dr. Rahatara Ferdousi

- Fewer generated tokens are a real saving only when the answer is still complete. **MinimalLM gives both, without training.** A guardrail instruction, an entropy-gated stop-token **logits processor**, a completeness gate and a final check make unchanged **Mistral-7B, Llama-3.1-8B and Qwen2.5-3B** generate **47.6% fewer tokens** on 330 questions (MT-Bench, AlpacaEval, Arena-Hard-Auto), with **no answer cut off**; with the same English instruction, **45.6%** fewer tokens on questions in Arabic, Chinese, French, German, Spanish and Russian.
- Completeness measured by a cut-off rule, an LLM judge (8.04/10 against 8.42) and NLI-based content coverage (RoBERTa-large-MNLI). Cost: the shorter answers keep 37% of the long baseline answer's content and more often leave out part of a broad question.

Horus-OSINT — Domain-Adapted Language Model for Event Intelligence *2026*
Meta-Llama-3-8B · Instruction Fine-Tuning · AWS · Python

- Curated and formatted **159,826** GTD/GDELT records into an instruction dataset, then fine-tuned **Meta-Llama-3-8B** for natural-language querying of global event data.
- Deployed the adapted model on **AWS** as an interactive conversational assistant over structured and unstructured intelligence sources.

HORUS Sentinel — Text-to-Knowledge-Graph Intelligence Pipeline *2026*
Information Extraction · Knowledge Graphs · Multi-Agent Systems · Fine-Tuned LLM

- Built a pipeline that turns continuously collected **unstructured open-source text** into a **living knowledge graph**, correlating entities and relations across sources rather than treating each document in isolation.
- A self-hosted fine-tuned model reasons over the graph to generate **prioritised, evidence-backed reports** with every claim traceable to source text.

Automated Multimodal Agent — Document Understanding to Narrated Output *2026*
Document Parsing · Text Generation · Speech Synthesis

- Built an autonomous pipeline that extracts and restructures document content into slide-level summaries, then renders them as narrated audio — chaining extraction, generation and synthesis end to end.

Additional Projects

- **SimCLR-Vision-SSL** — contrastive representation learning with **FAISS** embedding retrieval — the same vector-search machinery used in semantic search and RAG.
- **Human vs. Agentic Pull Requests** — large-scale text and metadata mining of GitHub pull-request discussions.
- **Also:** AudioShield (audio classification) · SCRAP · RL-Cloud-Autoscaler · Vitalism Solution (rPPG).

CERTIFICATIONS & AWARDS

AWS Academy — Machine Learning for NLP
Generative AI & Prompt Engineering — IBM / Coursera
HCIP – AI (Huawei)
1st Place — Smart Cities Hackathon 2022
AWS Academy — ML Foundations, Data Engineering
Machine Learning Specialization — DeepLearning.AI
Digilians Presidential Scholarship — Queen's University
Semi-Finalist — Microsoft Imagine Cup 2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency.